EduAirControl — General Project Document

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EduAirControl — General Project Document

> Overview, scope, architecture, organization and status of the project

National Learning Service (SENA) · Software Analysis and Development – 3145556

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Instructors: All program instructors

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1. General project information

This document gives a complete, high-level view of EduAirControl: the problem it solves, who it serves, what it includes and excludes, how it is built, who builds it, how the work is managed and what its current status is. It is the entry point to the rest of the project deliverables; each topic is covered in depth in the document listed in section 12.

• | Field | Detail |
• |---|---|
• | Project name | EduAirControl |

• | Type of solution | IoT-based school environmental monitoring system with a web application, a mobile application and a microservices backend |

• | Training program | Technologist in Software Analysis and Development – 3145556 |

• | Institution | National Learning Service – SENA |

• | Team members | María de los Ángeles Olaya García; Jhon Jairo Cortez Vargas |

• | Instructors | All program instructors |

• | Period | 2025-2026 |

• | Product status | MVP under development (unreleased version) |

• | Baseline documentation | SRS v1.1 (IEEE 830), updated on 2026-10-08 against the code and the engineering documentation |

• | Documentation repository | github.com/code-sena/ea-control-docs |

• | Version of this document | 1.0 – October 9, 2026 |

2. Introduction

Well-being in school environments has become a key factor for quality education. Environmental conditions inside the classroom directly affect the concentration, academic performance and health of students and teachers. In this context, technological solutions are needed to monitor and improve these spaces.

EduAirControl is a school environmental monitoring system based on IoT (Internet of Things) technology. It records the temperature, carbon dioxide (CO■), humidity and noise level of educational environments and presents them in a web application and a mobile application, together with severity-based alerts, historical analysis and recommendations to improve the measured conditions.

Besides optimizing the physical conditions of the classroom, the project seeks to foster a culture of environmental and technological awareness in the educational community, in line

with the principles of sustainability, health and digital transformation.

3. Problem statement

In many educational institutions, environmental conditions (temperature, CO■, humidity and noise) are not monitored continuously. These variables are invisible, yet they affect the concentration, learning and health of students and teachers. Without real-time information it is not possible to make timely decisions, such as ventilating a room when CO■ rises, and the measurements that do exist are scattered and lack a history that shows changes over time.

> Research question. How can continuous monitoring of environmental variables such as temperature, carbon dioxide, humidity and noise level in educational institutions improve the concentration, learning and health of students and teachers?

4. Purpose and justification

4.1 Purpose

To develop and implement a school environmental monitoring system that uses IoT technology to record the classroom's environmental variables in real time, improve the physical conditions of the educational setting and support, with data, decisions that strengthen the quality of learning.

4.2 Justification

According to the World Health Organization guidelines (2021), poor ventilation in enclosed spaces such as classrooms lets CO■ build up, causing drowsiness, fatigue and reduced attention. Satish et al. (2012) found negative effects on decision-making at CO■

concentrations above 1,000 ppm. The EPA (2020) also reports that exposure to high temperatures can reduce student performance and that excess noise increases the likelihood of errors in schoolwork.

In Colombia, most public educational institutions lack systems to monitor these variables. EduAirControl responds with an accessible solution: it collects and analyzes environmental data in real time, generates alerts and statistics useful for pedagogical and administrative decisions, and promotes technology adoption, aligning with the guidelines of the Ministry of National Education and the Sustainable Development Goals of the 2030 Agenda.

5. Objectives

5.1 General objective

To develop an IoT-based school environmental monitoring system that records and displays temperature, carbon dioxide (CO■), humidity and noise level in real time, in order to improve learning conditions and the well-being of students and teachers in educational environments.

5.2 Specific objectives

• Design a software system that integrates sensors for real-time monitoring of temperature, CO■, humidity and noise in the educational environments of one or several institutions.
• Develop an intuitive interface that lets administrators and teachers view environmental data and receive alerts when conditions fall outside the optimal ranges.
• Create an analysis module that processes the measurement history by day, week, month and year to identify patterns and support decision-making.

• Integrate a severity-based alert and notification system that informs those responsible about anomalies and enables immediate corrective action.

• Guarantee secure access, isolated by institution and campus, through standard OAuth2/OpenID Connect authentication and role-based access control.

6. Scope

6.1 Capabilities included in the MVP

• | # | Capability | Description | Responsible service |

• |---|---|---|---|

• | 1 | Authentication and registration | Sign-up, login (email and Google/Facebook), password recovery and role-based access control. | ms-security |

• | 2 | Institutions and campuses | Creation and management of institutions (tenants) and campuses; data isolation per institution. | ms-security, ms-classroom-management |

• | 3 | Educational environments | Creation, query, editing and deletion of environments with type, location and capacity. | ms-classroom-management |

• | 4 | Sensor management | Registration, installation/removal and configuration of sensors and the variables they measure. | ms-sensor-management |

• | 5 | Environmental monitoring | Reception and storage of measurements; current status of temperature, humidity, CO■ and noise. | ms-environment-monitoring |

• | 6 | Dashboard and historical analysis | Current and historical visualization by day, week, month and year. | ms-environment-monitoring, frontend |

• | 7 | Alerts and notifications | Threshold-based alerts (INFO, WARNING, CRITICAL) and per-user read status. | ms-environment-monitoring, ms-user-experience |

• | 8 | Favorites, ratings and preferences | Favorite environments, 1-to-5 rating with comment, language, theme and accessibility. | ms-user-experience |

• | 9 | Client applications | Web application (React) and mobile application (Expo/React Native) with BLE configuration of ESP32 devices. | frontend |

6.2 Out of scope for this version

• Advanced environmental prediction.

• Automatic equipment control (ventilation, air conditioning).

• Integration with external environmental control systems.

• Development of the sensor hardware (supplied by a provider or the institution).

6.3 Constraints and assumptions

• The application requires internet access and IoT devices must be powered on and connected to a Wi-Fi network.

• Base technology: React, Expo/React Native, Spring Boot, PostgreSQL 15, Redis, RabbitMQ and Docker.

• Work is organized in one-week sprints.

• The budget for sensors and microcontrollers is limited, which constrains the quantity and quality of the equipment.

• Each user belongs to a single institution and a single campus and cannot see data from other institutions.

• Stable Wi-Fi and adequate power supply are assumed in the spaces, and that the chosen sensors have acceptable accuracy for basic monitoring.

7. Product description

7.1 Users and roles

• | Role | Profile | What they do in the system |

• |---|---|---|

• | SUPER_ADMIN | Platform technical team | Creates and manages institutions; manages users and roles across the whole platform; handles errors and supports administrators. |

• | ADMIN | Person in charge of an institution, with office-software skills | Manages the environments, sensors, devices and thresholds of their institution. |

• | USER | Teachers, administrators and students | Views environments and variables, filters, marks favorites, rates environments, reviews alerts and sets preferences. |

7.2 Monitored variables

Variables and their ranges are defined once (ADR-014) and are used identically by the backend, the web and the mobile app. Normal, warning and critical thresholds are configured per environment type (classroom, laboratory and library).

• | Variable | Code | Unit | Valid range | Classroom example: normal |

• |---|---|---|---|---|

• | Temperature | TEMP | °C | -10 to 50 | 20 – 26 |

• | Humidity | HUM | % | 0 to 100 | 40 – 60 |

• | Carbon dioxide | CO2 | ppm | 0 to 5000 | 0 – 800 |

• | Noise level | NOISE | dB | 0 to 140 | 0 – 45 |

7.3 Main user journeys

• Authorized consultation: the user signs up with their institution code or logs in, picks an environment, sees its real-time metrics and queries the history by day, week, month or year.

• Administrative setup: the administrator creates the environment, registers the sensor, installs it in the environment, links the variables it measures and defines the thresholds.

• From measurement to decision: the sensor sends a measurement; the system validates it, stores it, compares it with the thresholds and raises an alert if needed; the user sees it on the dashboard and acts (for example, ventilates the room).

8. Architecture and technology

EduAirControl follows a microservices architecture with Spring Boot behind an API Gateway that acts as a BFF (Backend for Frontend). Each service has its own database schema (ADR-003), is organized with hexagonal architecture (ADR-009) and communicates with the others through REST or RabbitMQ events.

```
flowchart TD
  WEB["Web application<br/>React + Vite"] --> GW
  MOB["Mobile application<br/>Expo / React Native"] --> GW
  IOT["IoT device<br/>ESP32 + sensors"] --> GW
  GW["API Gateway (BFF)<br/>:8080 · OAuth2/OIDC + PKCE"]
  GW --> SEC["ms-security :8081"]
  GW --> CLS["ms-classroom-management :3002"]
  GW --> SEN["ms-sensor-management :3004"]
  GW --> MON["ms-environment-monitoring :3003"]
  GW --> UXP["ms-user-experience :3006"]
  GW --> USR["ms-user-management :3007"]
  SEC --> PG[("PostgreSQL 15")]
  MON --> PG
  GW --> RD[("Redis")]
  SEN --> MQ{{"RabbitMQ"}}
  MQ --> MON
```

Figure 1. Overview of EduAirControl components. Full detail is in 05_Diseno_Software.

8.1 Backend services

• | Service | Responsibility | Port | Status |

• |---|---|---|---|

• | api-gateway | Single entry point, authentication, routing and rate limiting (BFF). | 8080 | Active |

• | ms-security | Sign-up, login, OAuth2/OIDC, tokens, roles and institutions. | 8081 | In development |

• | ms-classroom-management | Educational environments, environment types and campuses. | 3002 | In development |

• | ms-environment-monitoring | Measurements, historical analysis, thresholds and alerts. | 3003 | Active |

• | ms-sensor-management | Sensors, variables and installations. | 3004 | In development |

• | ms-user-experience | Favorites, ratings, preferences, searches and notification status. | 3006 | Active |

• | ms-user-management | User information. | 3007 | In development |

• | ms-alert | Retired: its functions moved to ms-environment-monitoring (ADR-015). | — | Deprecated |

8.2 Technology stack

• | Layer | Technology | Role |

• |---|---|---|

• | Web frontend | React 19, Vite | Responsive web interface with light/dark mode and color-blindness themes |

• | Mobile app | Expo 54, React Native 0.81 | Android and iOS app with BLE device configuration |

• | Backend | Java 17, Spring Boot | Domain services and APIs under hexagonal architecture |

• | API entry | Spring Cloud Gateway | Routing, authentication, rate limiting and httpOnly cookie session |

• | Authentication | Spring Authorization Server, JWT RS256 | OAuth2/OIDC with Authorization Code + PKCE; Google and Facebook |

• | Data | PostgreSQL 15, Liquibase | Relational persistence and schema version control |

• | Support | Redis, RabbitMQ | Sessions and revoked tokens; events between services |

• | Contracts | OpenAPI | One contract per service in 07-api/contracts/openapi |

• | DevOps | Docker, Jenkins, GitHub | Containers, continuous integration and source control |

• | Observability | Prometheus, Grafana, OpenTelemetry | Metrics, dashboards and traces (defined in ADR-012) |

8.3 IoT hardware

The measurement prototype is based on an ESP32 microcontroller with low-cost sensors. The mobile app configures the device's Wi-Fi network over Bluetooth Low Energy (the device advertises itself as ESP32_Config).

• | Component | Function |

• |---|---|

• | ESP32 | Controller and Wi-Fi/BLE communication |

• | DHT11 | Air temperature and humidity |

• | KY-038 | Relative sound level |

• | MQ-135 | Gases and air quality; not a precision CO■ meter and its voltage must be adapted before the ADC |

> Current hardware status. Test programs exist for the DHT11 and the KY-038. The final firmware that sends readings to the API, and the definitive MQ-135 wiring, are still pending confirmation. Therefore the system can receive measurements over HTTPS/MQTT, but the complete sensor → API chain is not considered closed in this deliverable.

9. Team and work organization

9.1 Team

• | Member | Role | Responsibility | Contact |

• |---|---|---|---|

• | María de los Ángeles Olaya García | Fullstack developer | Information analysis, design and programming | mariadelosangelesolayagar@gmail.com |

• | Jhon Jairo Cortez Vargas | Fullstack developer and tech lead | Information analysis, design and programming; technical decisions | cortezvargasjhon01@gmail.com |

9.2 Methodology

The team works with an agile approach based on one-week sprints and spec-driven documentation: the requirement, decision or contract is documented first, and then built. The practices applied in the repository are:

• Documented governance: agile conventions, definition of ready (DoR) and of done (DoD), Git conventions, documentation rules and security policies (00-governance folder).

• Architecture decisions: 17 numbered ADRs (ADR-001 to ADR-017) that record the reasoning behind each decision.

• Branches by type of work: feat/, fix/, docs/ and chore/, merged through pull requests with a template.

• Traceability: user stories, a traceability matrix and functional and non-functional requirements linked to each other (04-requirements folder).

• Contracts first: each microservice publishes its OpenAPI contract before being integrated.

10. Current status and planning

10.1 Requirements status

According to SRS v1.1 (updated on 2026-10-08), the 40 functional requirements stand as follows:

• | Status | Count | Requirements |

• |---|---|---|

• | Implemented | 26 | RF1.1, 1.3–1.9; RF2.1, 2.4; RF3.1, 3.2; RF5.1; RF6.1, 6.2, 6.4; RF7.1–7.5; RF8.1; RF9.1–9.4 |

• | Partial | 12 | RF2.2, 2.3; RF3.3, 3.4; RF4.1, 4.2; RF5.2, 5.3; RF6.3; RF8.2, 8.3; RF9.5 |

• | Pending | 2 | RF1.2 (two-step verification); RF6.5 (external push or email notification) |

10.2 Project milestones

• | Date | Milestone |

• |---|---|

• | 2025 | Initial version of the SRS (IEEE 830). |

• | 2026-09-03 | Start of the engineering documentation repository (Docs-ea-control). |

• | 2026-09-15 / 17 | Refinement of product, requirements and architecture; UX/UI documentation and web and mobile mockups. |

• | 2026-10-05 | Analysis contract of ms-environment-monitoring. |

• | 2026-10-08 | Source-code audit, risk register and SRS v1.1 aligned with the implementation. |

• | 2026-10-09 | General document and consolidation of the project deliverables. |

10.3 Roadmap

• | Horizon | Focus | Content | Exit condition |

• |---|---|---|---|

• | H1 – Now | Core MVP | Sign-up and authentication, environment and sensor management, measurement ingestion, dashboard and historical analysis, favorites and ratings. | A representative user completes the authorized consultation journey with valid data. |
• | H2 – Next | Validated iteration | Improve dashboard clarity, filters, preferences and searches based on pilot evidence; strengthen data completeness and operational visibility. | Usage evidence identifies concrete improvements without leaving the product's domain. |
• | H3 – Later | Controlled expansion | External notifications, richer operational analysis or other integrations, only with approved value, data ownership and contracts. | Each expansion has explicit requirements, data, API and architecture review. |

11. Main risks

The risk register was updated from the code audit of 2026-10-08. The active risks are summarized below; their mitigation plans and the corrective, preventive and improvement actions are detailed in the 08_Calidad_Mejoras document.

• | ID | Risk | Level |
• |---|---|---|
• | R-001 | Identity spoofing through the X-User- headers, which services accept without verifying their origin. | Critical |
• | R-002 | RSA private key committed in the ms-security repository. | Critical |
• | R-003 | Open CORS with credentials on the gateway. | High |
• | R-004 | Test coverage not enforced in continuous integration, despite NFR6. | High |
• | R-005 | ms-user-experience missing from the Kubernetes deployment and from continuous integration. | High |
• | R-006 | Observability documented but not effective in the code. | High |
• | R-007 | Gap between ADR-003 (database per service) and the actual deployment. | Medium |

• | R-008 | Circuit breaker documented (ADR-013) but not implemented. | Medium |

• | R-009 | Device firmware outside the API contract. | Medium |

• | R-010 | Frontend without tests or shared contracts. | Medium |

• | R-011 | Documentation describing elements that do not exist yet. | Medium |

12. Project deliverables

The project is delivered in a folder with the following organization. This document is the first file; the others are built on the same basis and in the same format. Every document is delivered as PDF and Markdown, in Spanish and English.

• | Deliverable | Content |

• |---|---|

• | 01_Documento_General_Proyecto | General project overview (this document). |

• | 02_Especificacion_Requisitos | SRS v1.1: functional and non-functional requirements, thresholds and use cases. |

• | 03_Analisis_Software | Software analysis and dynamic views. |

• | 04_Propuesta_Tecnica | Technical proposal, decisions and technology rationale. |

• | 05_Diseno_Software | Graphical interfaces, architectural model, design patterns and data model. |

• | 06_Base_Datos | Entity-relationship model, data dictionary and SQL script. |

• | 07_Pruebas | Test plan and evidence. |

• | 08_Calidad_Mejoras | Corrective, preventive and improvement actions. |

• | 09_Implementacion | Software implementation procedure. |

• | 10_Manual_Tecnico · 11_Manual_Instalacion · 12_Manual_Usuario | Technical, installation and end-user manuals. |

• | 13_Articulo_Proyecto | Project article. |

• | 14_Anexos · 15_Enlace_Repositorio_GitHub | Supporting material and link to the code repository. |

13. Glossary

• | Term | Description |

• |---|---|

• | Educational environment | Space (classroom, laboratory, library) monitored by EduAirControl. |

• | Institution (tenant) | Educational organization using the platform; its data is isolated from that of others. |

• | Campus | Subdivision of an institution that contains environments and sensors. |

• | IoT | Internet of Things: network of connected sensors and microcontrollers. |

• | ESP32 | Microcontroller with Wi-Fi and Bluetooth used in the measurement devices. |

• | BLE | Bluetooth Low Energy; used to configure the device from the mobile app. |

• | OAuth2 / OIDC / PKCE | Authorization and identity standards used for login. |

• | BFF | Backend for Frontend: the gateway stores the tokens and gives the browser only an httpOnly cookie. |

• | RBAC | Role-based access control (USER, ADMIN, SUPER_ADMIN). |

• | ADR | Architecture decision record. |

• | SRS | Software Requirements Specification. |

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